

MUHAMMAD HAZIQ HASBI BIN OSMAN

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SUMMARY

Results-driven Electronic Engineering graduate with practical experience in semiconductor manufacturing, machine maintenance, electronic component testing, and network system installation. Completed internships at Robert Bosch Semiconductor Manufacturing and SSIS E Solution Sdn. Bhd. Skilled in soldering, troubleshooting, spare part management, machine buy-off support, and workpiece carrier repair. Experienced in using Microsoft Excel for data tracking and operational monitoring. Possesses strong understanding of industrial systems, equipment reliability, semiconductor processes, and electronic hardware functionality, with a commitment to quality, continuous improvement, and engineering excellence.

EDUCATION

Bachelor's Degree with Honours in Electronic Engineering

Universiti Tun Hussein Onn Malaysia (UTHM), Batu Pahat, Johor
September 2021 – July 2025

Diploma in Electrical & Electronic Engineering

Politeknik Seberang Perai, Permatang Pauh, Penang
June 2018 – August 2021

ACADEMIC PROJECTS

Prediction for Maternal Health Using Tree-Based Machine Learning Model (2025)

- Developed a machine learning model to predict maternal health risk levels based on clinical attributes.
- Applied data analysis and predictive modeling techniques to improve classification accuracy.

Energy Consumption Monitoring System with IoT (2024)

- Designed and developed an IoT-based energy monitoring system for residential and industrial environments.
- Enabled real-time energy tracking and remote monitoring.

Autonomous Fire Fighter Robot (2021)

- Developed an autonomous robot capable of detecting and extinguishing indoor fires.
- Integrated sensors and autonomous navigation mechanisms for fire suppression.

CERTIFICATES & TRAINING

IC Design Training Program: Design Verification – PSDC (2025)

- Hands-on experience with SystemVerilog and Verilog testbench development.
- Performed simulation execution, waveform analysis, and debugging in Linux-based environments.
- Learned UVM concepts, SystemVerilog Assertions (SVA), and CMOS fundamentals.

Additional Certifications:

- Integrated Design Project (IDP) Showcase – Silver Award (2024)
- MATLAB Calculus & Ordinary Differential Equation Training (2024)
- Dean’s List Award – Semester 2 & 7

PROFESSIONAL EXPERIENCE

Degree Internship – Robert Bosch Semiconductor Manufacturing Penang Sdn. Bhd. Batu Kawan, Penang | July 2024 – September 2024

- Managed and monitored spare part inventory for semiconductor equipment.
- Developed an Excel-based spare parts tracking system to improve data accuracy.
- Supported machine buy-off activities and documentation review.
- Conducted testing and repair of workpiece carriers.
- Performed soldering and component replacement on PCB assemblies.

Diploma Internship – SSIS E Solution Sdn. Bhd. Bukit Mertajam, Penang | March 2021 – July 2021

- Assisted in installation and configuration of fiber optic networks and server racks.
- Performed Ethernet cable termination and LAN infrastructure deployment.
- Conducted troubleshooting and maintained technical documentation.

SKILLS

Software & Tools:

Microsoft Excel, MATLAB, Arduino IDE, Proteus, Vivado, Visual Studio Code, Linux

Programming & Technical Skills:

SystemVerilog, Verilog HDL, UVM Basics, SystemVerilog Assertions (SVA), C++, Python, PCB Soldering, Preventive Maintenance, Troubleshooting

LANGUAGES

- Bahasa Melayu
- English

REFERENCES

Dr. Nur Ilyani Binti Ramli – Supervisor Final Year Project | UTHM

Anbarasu Balasubramaniam – Supervisor | Robert Bosch Semiconductor Manufacturing

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